8.14 ECC Mixer Component, ECCMIX

An ECC mixer is indicated by ECCMIX for Word 2 on card CCC0000. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N is the face number.

An ECC mixer (ECCMIX) component is a specialized branch that requires three junctions with a certain numbering order. The physical extent of the ECC mixer is a length of the cold leg, or any other horizontal pipe, centered around the position of the ECC injection location. The length of this pipe segment should be equal to three times the inside diameter of the pipe (if the physical arrangement of the system permits). Junction number 1 is the ECC connection. Junction number 2 is the flow inlet to the ECC mixer component in normal operation. The geometrical angle between the axis of junctions 1 and 2 is input on the ECC Mixer Junction Geometry card for junction number 1. Junction number 3 is the discharge, junction in normal operation. The to part of junctions 1 and 2 must refer to the inlet end of the ECC mixer (CCC010001), and the from part of the discharge junction must refer to the outlet end of the ECC mixer (CCC010002). Note that junction 1 is not a crossflow junction. Two or more ECCMIX components may be considered in modeling some piping. These may be connected in tandem and require at least one normal volume between them.

8.14.1 ECC Mixer Component Information, CCC0001

This card is required.

W1(I) Number of junctions, nj. This number must be 3 for ECCMIX components.

W2(I) Initial condition control. This word is optional and, if missing, the junction initial velocities in the first and second words on card CCCN201 are assumed to be velocities. If 0, velocities are assumed; if nonzero, mass flows are assumed.

8.14.2 ECC Mixer X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m^2, ft^2).

W2(R) Length (m, ft).

W3(R) Volume (m^3, ft^3). The code requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be $\leq$ 360 degrees. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). The value of this angle must be less than $\pm$ 15 degrees. Any other value will be considered an input error. The angle 0 degrees is horizontal, and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. For ECCMIX, the ECC mixer flow regimes are used.

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \left(\frac{\text{area}}{\text{wetted perimeter}}\right)$.

If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. A check is made that the pipe roughness is less than half the hydraulic diameter. See Word 1 for the area.

W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag. For the ECCMIX component, the volume control flags are limited to a format of 00p00fe.

p-flag The digit p specifies whether the water packing scheme is to be used. p = 0 specifies that the water packing scheme is to be used for the volume, and p = 1 specifies that the water packing scheme is not to be used for the volume.

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the x-coordinate direction in the volume, and f = 1 specifies that wall friction effects are not to be computed along the x-coordinate.

e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison to other codes.

8.14.3 ECC Mixer Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

8.14.4 ECC Mixer Volume Initial Conditions, CCC0200

This card is required.

W1(I) Control word. This word has the format εbt. It is not necessary to input leading zeros.

e-flag The digit ε specifies the fluid, where ε = 0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-129 cards (i.e., ε = 1 specifies H_2O, ε = 2 specifies D_2O etc.). The default fluid is that set for the hydrodynamic system by cards 120-129 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-129 or these control words must be consistent (i.e., not specify different fluids). If cards 120-129 are not entered and all control words use the default ε = 0, then water is assumed to be the fluid. It is recommended that the user use the 120-129 cards to set the fluid type in the systems.

b-flag The digit b specifies whether boron is present. b = 0 specifies that the volume liquid does not contain boron, and b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit t specifies how the following words are to be used to determine the initial thermodynamic state. t = 0-3 specifies one component (steam/water); t = 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If t = 0, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If t = 1, the next two words are interpreted as temperature (K, ^oF) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If t = 2, the next two words are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If t = 3, the next two words are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, ^oF) in equilibrium condition. Enter boron in W4 if needed.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0E-9 \leq x_s \leq 0.99999999$,

Single phase if $x_s < 1.0\ E-9$ or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If t = 4, the next three words are interpreted as pressure (Pa, $lb_f/in.^2$), temperature (K, ^oF), and static quality in equilibrium condition. Using this input option with static quality greater than 0.0 and less than or equal to 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option. Enter boron in W5 if needed.

[T,x_s,x_n] If t = 5, the next three words are interpreted as steam saturation temperature (K, ^oF), static quality, and noncondensable quality in equilibrium condition. Enter boron in W5 if needed. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

[P,U_f,U_g,α_g,x_n] If t = 6, the next five words are interpreted as pressure (Pa, $lb_f/in.^2$), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing (t = 0). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor-specific internal energy. Enter boron in W7 if needed.

W2-W7(R) Quantities as described under Word 1. Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass of liquid) follows the last required word for thermodynamic conditions.

8.14.5 ECC Mixer Junction Geometry, CCCN101-N109

These cards are required. Cards with N equal to 1, 2, and 3 are entered, one for each junction. For the first junction, the ECC injection connection, enter 7 words, and for junctions 2 and 3, enter 6 words.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2, ft^2). If 0, the area is set to the minimum area of the adjoining volumes. For abrupt area changes, the junction area must be equal to or smaller than the minimum of the adjoining areas. For smooth area changes, there are no restrictions.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.10.10). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in W3. If one of the abrupt area change options is selected (a = 1 or 2), the entry here is the loss coefficient appropriate for the minimum flow area between the two hydrodynamic volumes connected by this junction. If the a = 1 option is selected, a loss coefficient entered here is considered to be additive with the abrupt area change flow energy loss coefficient, which is calculated separately by the code.

W5(R) Reynolds number independent reverse flow energy loss coefficient, A_R. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is negative. A variable loss coefficient may be specified (see Section 8.10.10). See information for the previous word regarding interpretation and use of this loss coefficient.

W6(I) Junction control flags. This word has the format jefvcahs. For the ECCMIX component, the format is restricted to 0000cah0.

c-flag The digit c specifies choking options. c = 0 means that the choking model will be applied, and c = 1 means that the choking model will not be applied. The choking model used is based on input on card 1. If card 1 is missing or it does not contain Option 50, the Henry-Fauske critical flow model is active. If card 1 is present and it contains Option 50, the original RELAP5 critical flow model is active.

a-flag The digit a specifies area change options. a = 0 means a smooth area change and only the user supplied K_{loss} is applied. a = 1 means full abrupt area change model and the code calculates forward and reverse contraction/expansion K_{loss} terms and adds them to the user supplied K_{loss} terms. In addition, the code includes extra interphase drag to account for the vena contracta. a = 2 means a partial abrupt area change model and it is the same as

¨a = 1 except no code calculated forward and reverse contraction/expansion K_{loss} is applied.

h-flag The digit h specifies nonhomogeneous or homogeneous. h = 0 specifies the nonhomogeneous (two-velocity momentum equations) option and h = 1 or 2 specifies the homogeneous (single-velocity momentum equation) option. For the homogeneous option (h = 1 or 2), the major edit printout will show h = 1.

W7(R) Angle (degrees) between the axis of the ECC injection line and the main pipe (or the angle between Junctions 1 and 2). This angle must be between 0 and 180 degrees. If missing, a 90-degree connection for the ECC pipe is assumed. Enter W7 only for the first junction.

8.14.6 ECC Mixer Junction Form Loss Data, CCCN112

These cards are optional. The values of N should follow the same approach as used in cards CCCN101-N109. The user-specified form loss is given in Words 4 and 5 of cards CCCN101-N109 if these cards are not entered. If these cards are entered, the form loss coefficient is calculated from

$$K_F = A_F + B_F Re^{-C_F}$$

$$K_R = A_R + B_R Re^{-C_R}$$

where K_F and K_R are the forward and reverse form loss coefficient. A_F and A_R are the Words 4 and 5 of cards CCCN101-N109. Re is the Reynolds number based on mixture fluid properties. If these cards are being used for the form loss calculation, then enter all four words for the appropriate expression. See the discussion regarding the flow area basis assumed for A_F and A_R; the same information applies for B_F and B_R.

W1(R) B_F (≥ 0). This quantity must be greater than or equal to 0.

W2(R) C_F (≥ 0). This quantity must be greater than or equal to 0.

W3(R) B_R (≥ 0). This quantity must be greater than or equal to 0.

W4(R) C_R (≥ 0). This quantity must be greater than or equal to 0.

8.14.7 ECC Mixer Junction Initial Conditions, CCCN201

Three cards are required. The values of N should be 1, 2, and 3.

W1(R) Initial liquid or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W2(R) Initial vapor velocity or mass flow (velocity in m/s, ft/s or mass flow in kg/s, lb/s).

W3(R) Interface velocity (m/s, ft/s). Enter 0.

8.15 Valve Component, VALVE

A valve junction component is indicated by VALVE for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the junction in the valve junction component is numbered CCC000000.

8.15.1 Valve Junction Geometry, CCC0101-0109

This card (or cards) is required for valve junction components.

W1(I) From connection code to a component. This refers to the component from which the junction coordinate direction originates. The connection code is CCCVV000N, where CCC is the component number, VV is the volume number, and N indicates the face number. The number N equal to 1 and 2 specifies the inlet and outlet faces respectively for the volume's coordinate direction. The number N equal to 3-6 specifies crossflow. The number N equal to 3 and 4 would specify inlet and outlet faces for the second coordinate direction; N equal to 5 and 6 would do the same for the third coordinate direction. For connecting to a time-dependent volume, N is restricted to 1 or 2.

W2(I) To connection code to a component. This refers to the component at which the junction coordinate direction ends. See the description for W1 above.

W3(R) Area (m^2, ft^2). This quantity is the full open area of the valve except in the case of a relief valve. For valves other than relief valves, if this area is input as 0, the area is set to the minimum area of adjoining volumes. If nonzero, this area is used. For relief valves, this term is the valve inlet throat area. If this term is input as 0, it will default to the area calculated from the inlet diameter term input on cards CCC0301-0309, in which case the inlet diameter term cannot be input as 0. If both this area and the inlet diameter are input as nonzero, this area will be used but must agree with the area calculated from the inlet diameter within 10^{-5} m^2. However, if this area is input as nonzero and the inlet diameter is input as 0, the inlet diameter will default to the diameter calculated from this area. When an abrupt area change model is specified, the area must be less than or equal to the minimum of the adjoining areas.

W4(R) Reynolds number independent forward flow energy loss coefficient, A_F. This quantity will be used in each of the phasic momentum equations when the junction velocity of that phase is positive or 0. A variable loss coefficient may be specified (see Section 8.15.3). The interpretation and use of this loss coefficient vary based upon the "a" junction control flag selected in W6. If the smooth area change option is selected (a = 0), the entry here is the loss coefficient appropriate for the junction flow area resulting from the entry made in